

Advanced topics in seasonal adjustment

JDemetra+: Overview of key capabilities

Jan Gerhardt, Christiane Hofer / DG Data and Statistics
CIC Seminar , 3-7 November 2025

■ Agenda

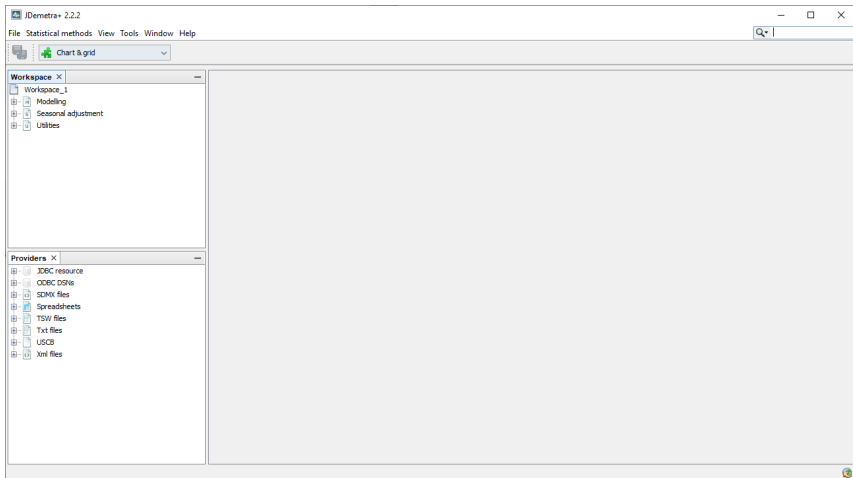
- Basic functionalities in JD+
- Enhanced functionalities via plug-ins
- Summary and useful links

In this section the following questions will be answered:

- How to import time series/data into JD+?
- How to carry out a seasonal adjustment (SA) in JD+?
- How to set or modify SA specifications?
- How to save and exchange workspaces/data/settings?

Basic functionalities

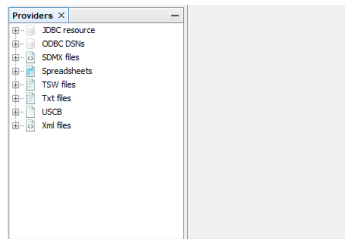
Starting the application for the first time



Basic functionalities

Data import (I/IV)

Providers Node structure of supported data sources



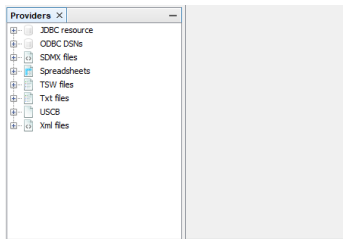
Basic functionalities

Data import (I/IV)

Providers Node structure of supported data sources

Data can be imported among other providers via

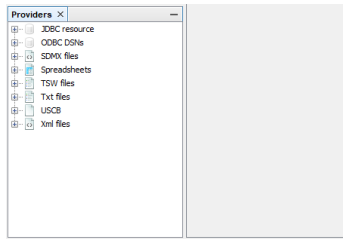
- “**Spreadsheets**” from Excel files (*.xls, *.xlsx)
- “**Txt files**” from Text files (*.txt, *.csv, *.tsv)
- “**SDMX**” from SDMX files (*.xml)
- “**ODBC DSNs**”, “**JDBC resource**” from standard databases (Oracle, SQLServer, DB2, etc.)
- plug-ins from further providers



Basic functionalities

Data import (II/IV)

How to import data:

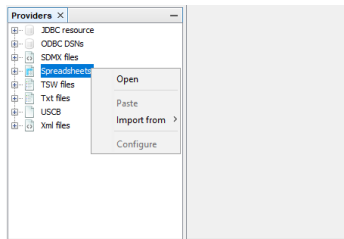


Basic functionalities

Data import (II/IV)

How to import data:

- Right-click on the selected provider
- Select **Open** in the context menu

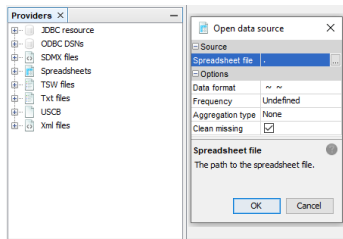


Basic functionalities

Data import (II/IV)

How to import data:

- Right-click on the selected provider
- Select **Open** in the context menu
- A new dialogue window pops up

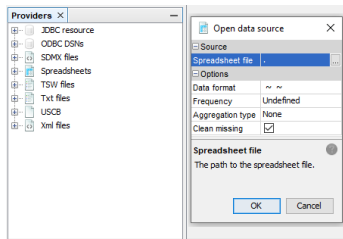


Basic functionalities

Data import (II/IV)

How to import data:

- Right-click on the selected provider
- Select **Open** in the context menu
- A new dialogue window pops up
- **Necessary:** Insert the path to the spreadsheet file.
- **Optional:** Set data formats (locale, date pattern, number pattern), frequency of the time series in **JD+**, handling of missing values

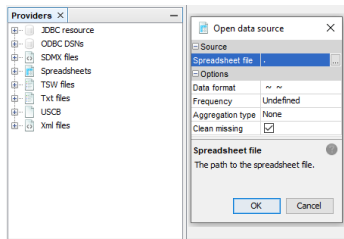


Basic functionalities

Data import (II/IV)

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- Select **Open** in the context menu
- A new dialogue window pops up
- **Necessary:** Insert the path to the spreadsheet file.
- **Optional:** Set data formats (locale, date pattern, number pattern), frequency of the time series in **JD+**, handling of missing values
- If the original time series have a higher frequency, you decide how these will be aggregated to the imported ones'



Basic functionalities

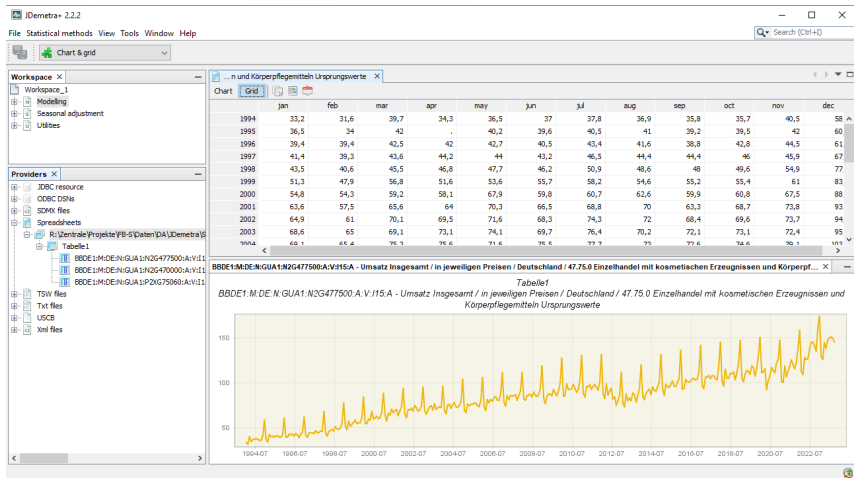
Functionalities (I/II)

Some nice features

- Customize the window to your own needs
- Add star: Load time series automatically
- Zoom in charts: Move mouse right down while pressed and other direction to reset
- Pressing key "R" highlights data points in a chart
- Drag and drop for time series
- Spreadsheet Profiler: Display the original data from the provider

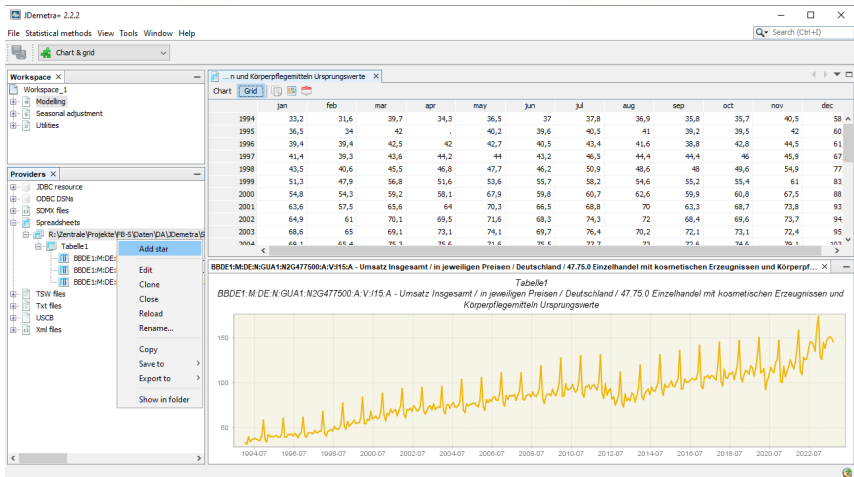
Basic functionalities

Functionalities (II/II)



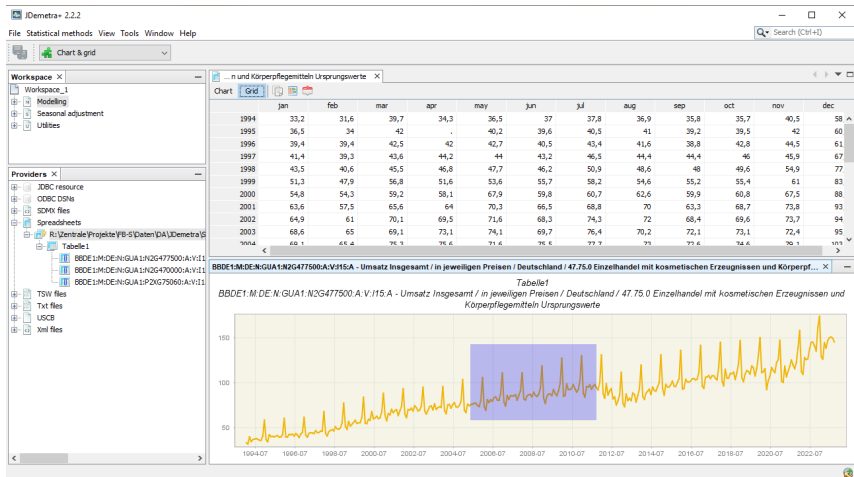
Basic functionalities

Functionalities (II/II)



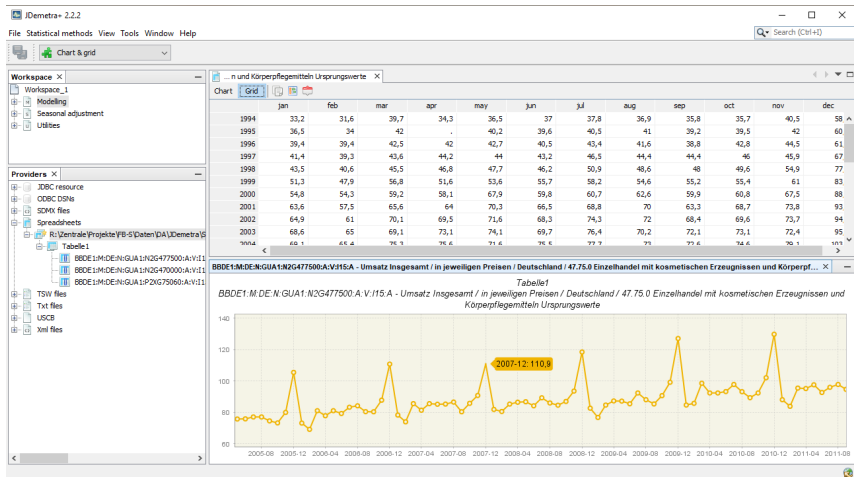
Basic functionalities

Functionalities (II/II)



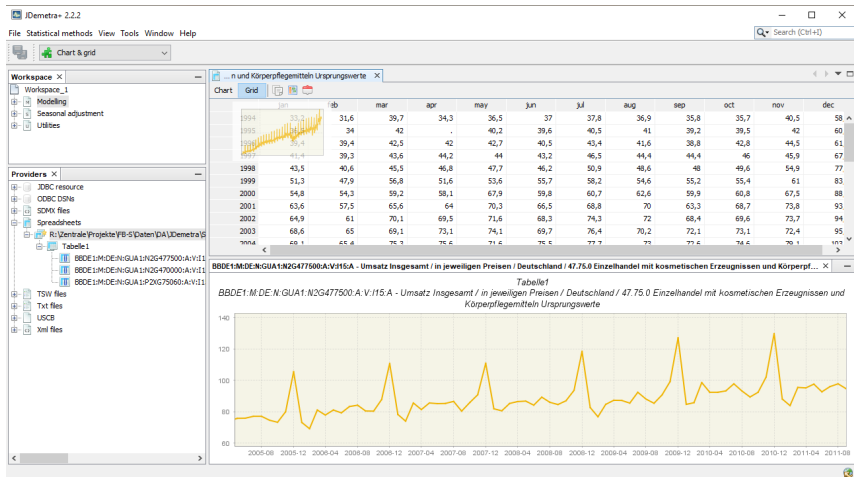
Basic functionalities

Functionalities (II/II)



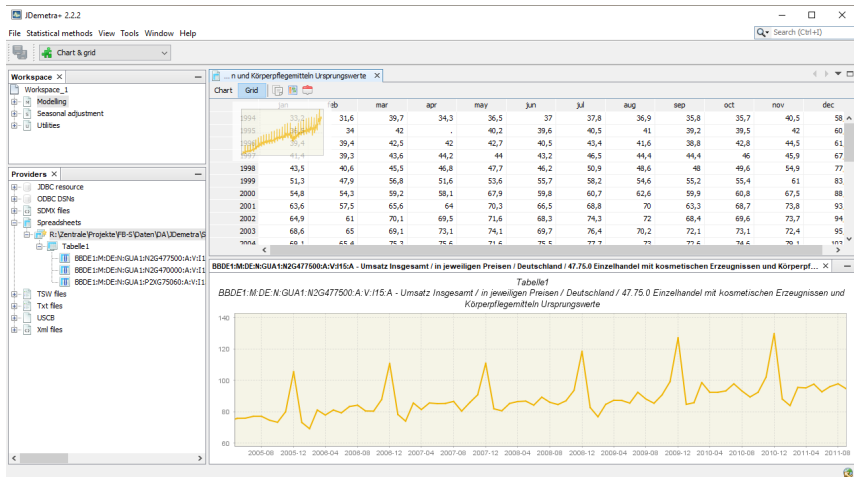
Basic functionalities

Functionalities (II/II)



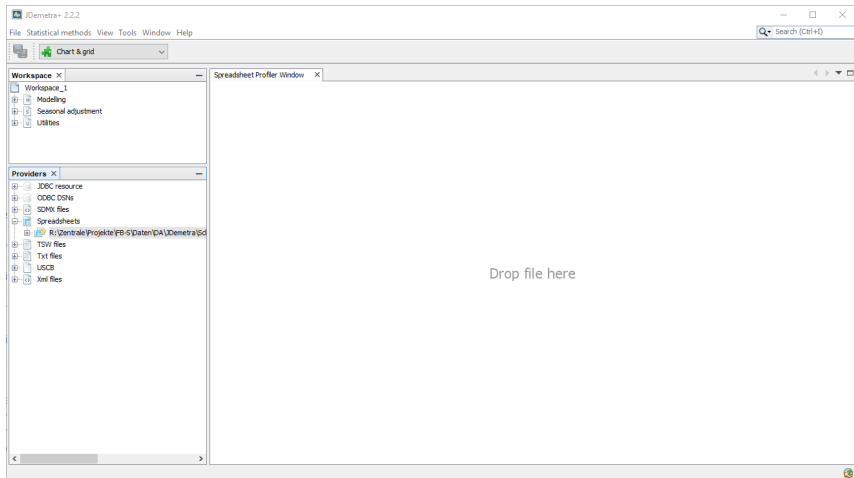
Basic functionalities

Functionalities (II/II)



Basic functionalities

Spreadsheet Profiler: Display original data



Basic functionalities

Spreadsheet Profiler: Display original data

The screenshot shows the iDemetra+ 2.2.2 application window. The main area displays a 'Spreadsheet Profiler Window' with a table named 'Tabelle1'. The table has four columns: A, B, C, and D. Column A contains dates from 1994-01-01 to 1996-02-01. Column B contains numerical values ranging from 33.2 to 39.4. Column C contains numerical values ranging from 74.9 to 77.4. Column D contains numerical values ranging from 83.6 to 133.4. The table is color-coded: rows 1-16 are red, rows 17-27 are green, and row 18 has a blue background. The word 'forty' is written in blue text in cell C17. The left sidebar shows a 'Workspace' with 'Workspace_1' and 'Providers' with various file types like JDBC resource, ODBC DSNs, SDMX files, Spreadsheets, TSV files, Text files, USCB, and Xml files. The top menu bar includes 'File', 'Statistical methods', 'View', 'Tools', 'Window', and 'Help'. A search bar is located in the top right corner.

	A	B	C	D
1		BBDE1:M...	BBDE1:M...	BBDE1:M...
2	1994-01-01	33.2	74.9	83.6
3	1994-02-01	31.6	73.9	75.3
4	1994-03-01	39.7	90	104
5	1994-04-01	34.3	80.1	98.2
6	1994-05-01	36.5	81.2	96.3
7	1994-06-01	37	79.7	88.2
8	1994-07-01	37.8	79.7	88.2
9	1994-08-01	36.9	79.6	84.8
10	1994-09-01	35.8	83.5	109
11	1994-10-01	35.7	85.2	114.1
12	1994-11-01	40.5	89.8	105.2
13	1994-12-01	38.3	109.9	137.7
14	1995-01-01	36.5	77.1	86.6
15	1995-02-01	34	75.6	78.2
16	1995-03-01	42	87.5	93.2
17	1995-04-01	forty	83.1	99.7
18	1995-05-01	40.2	86.1	101.4
19	1995-06-01	39.6	81.6	89
20	1995-07-01	40.5	81	90.5
21	1995-08-01	41	82.1	87.7
22	1995-09-01	39.2	83.8	109
23	1995-10-01	39.5	85	106.3
24	1995-11-01	42	91.6	112.4
25	1995-12-01	60.3	107	133.4
26	1996-01-01	39.4	78.7	84.9
27	1996-02-01	39.4	77.4	73.2

Basic functionalities

Spreadsheet Profiler: Display original data

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25	1995-12-01	60.3	107	133.4
26	1996-01-01	39.4	78.7	84.9
27	1996-02-01	39.4	77.4	73.2

Exercise: Providers

Import data from the exchange directory

R:/Zentrale/3/TZK/Daten/INT_SeasonalAdjustment/02Tuesday/JD+KeyCap

- From the file *ZR_EH.xlsx*
- Download the SDMX file (SDMX-ML 2.0) for the time series *BBDE1.M.DE.W.GUA1.N2G470000.A.V.I15.A* from the Bundesbank's time series databases and import it into JD+.

Basic functionalities

Seasonal Adjustment (I/III)

JDemetra+ offers two possibilities to conduct a seasonal adjustment:

Single Analysis:

- One time series
- One specification
- Single document

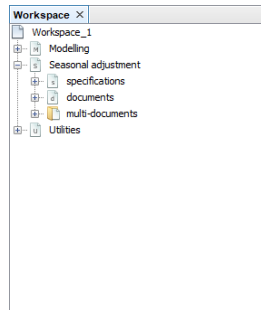
Multi-processing:

- Several time series with individual specifications
- Multi document

Basic functionalities

Seasonal Adjustment in JD+ (II/III)

Workspace Node structure containing settings for seasonal adjustment

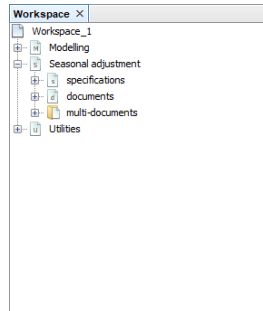


Basic functionalities

Seasonal Adjustment in JD+ (II/III)

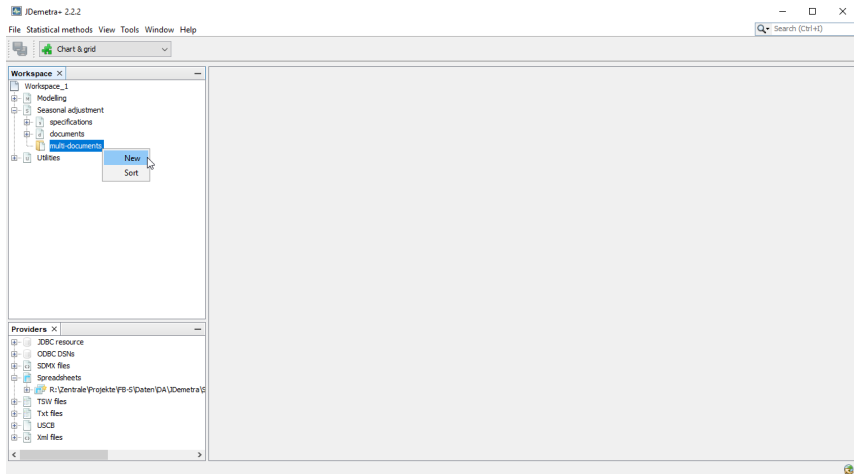
Workspace Node structure containing settings for seasonal adjustment

- **Modelling** find an ARIMA model for a time series
- **Seasonal adjustment: specifications** both pre-defined and individually customized
- **Seasonal adjustment: documents** Single Analysis of one time series with one specification
- **Seasonal adjustment: multi-documents** Multi-processing of several time series with individual specifications
- **Utilities** calendars, user-defined variables



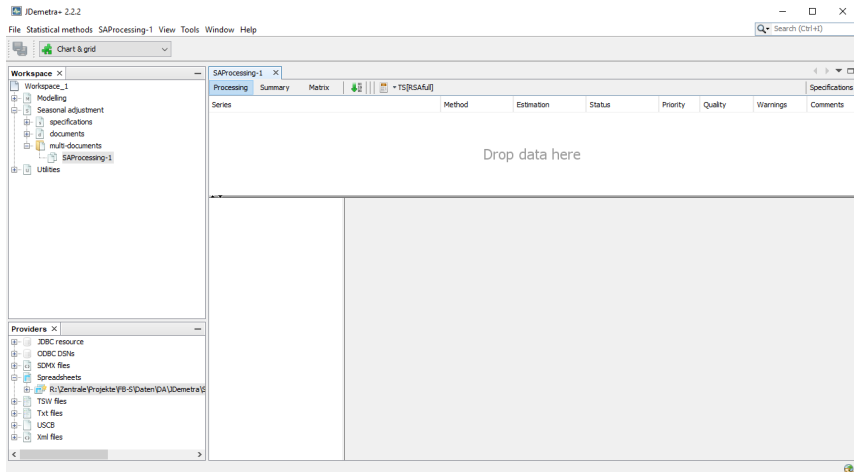
Basic functionalities

Seasonal Adjustment in JD+ (III/III)



Basic functionalities

Seasonal Adjustment in JD+ (III/III)



Basic functionalities

Seasonal Adjustment in JD+ (III/III)

JDemetra- 2.2.2

File Statistical methods View Tools Window Help

Search (Ctrl+F)

Chart & grid

Workspace X

Workspace_1

- Modeling
 - Seasonal adjustment
 - specifications
 - documents
 - multi-documents
 - SAPProcessing-1
 - Utilities

Providers X

- JDBC resource
- ODBC DSNs
- ODMX files
- Spreadsheets
 - R:\Zentrale Projekte\FB-S\Daten\DA\JDemetra\...
- TSV files
- Txt files
- USCB
- Xml files

SAPProcessing-1 X

Processing Summary Matrix  Items TS[RSaful]

Series	Method	Estimation	Status	Priority	Quality	Warnings	Comments
Tabelle1 ► BBDE1:M:DE:N:GJA1:N2G477500:A:V:I15:A - Umsa...	RSaful	Concurrent	Unprocessed				
Tabelle1 ► BBDE1:M:DE:N:GJA1:N2G470000:A:V:I15:A - Umsa...	RSaful	Concurrent	Unprocessed				
Tabelle1 ► BBDE1:M:DE:N:GJA1:P2XG75060:A:V:I15:A - Umsa...	RSaful	Concurrent	Unprocessed				

Basic functionalities

Seasonal Adjustment in JD+ (III/III)

iDemetra+ 2.2.2

File Statistical methods SAProcessing-1 View Tools Window Help

Search (Ctrl+F)

Chart & grid

Workspace X

Workspace_1

- Modeling
- Seasonal adjustment
 - specifications
 - documents
 - multi-documents
 - SAProcessing-1
 - Utilities

Providers X

- JDBC resource
- ODBC DSNs
- SOMX files
- Spreadsheets
 - R:\Zentrale Projekte\FB-S\Daten\DA\iDemetraS
- TSV files
- Txt files
- USCB
- Xml files

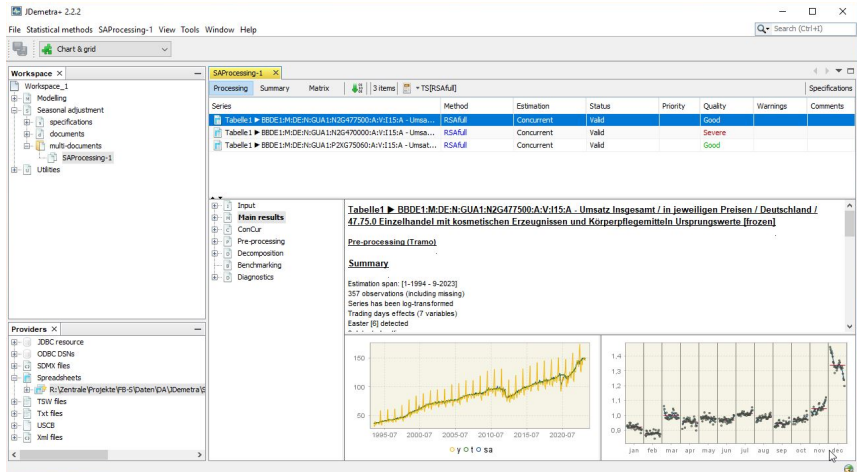
SAProcessing-1 X

Processing Summary Matrix 3 items TS[RSaful]

Series	Method	Estimation	Status	Priority	Quality	Warnings	Comments
Tabelle 1 ► BBDE1:M:DE:N:GUA1:N2G477500:A:V:I15:A - Umsa...	RSaful	Concurrent	Valid		Good		
Tabelle 1 ► BBDE1:M:DE:N:GUA1:N2G470000:A:V:I15:A - Umsa...	RSaful	Concurrent	Valid		Severe		
Tabelle 1 ► BBDE1:M:DE:N:GUA1:P2XG75060:A:V:I15:A - Umsa...	RSaful	Concurrent	Valid		Good		

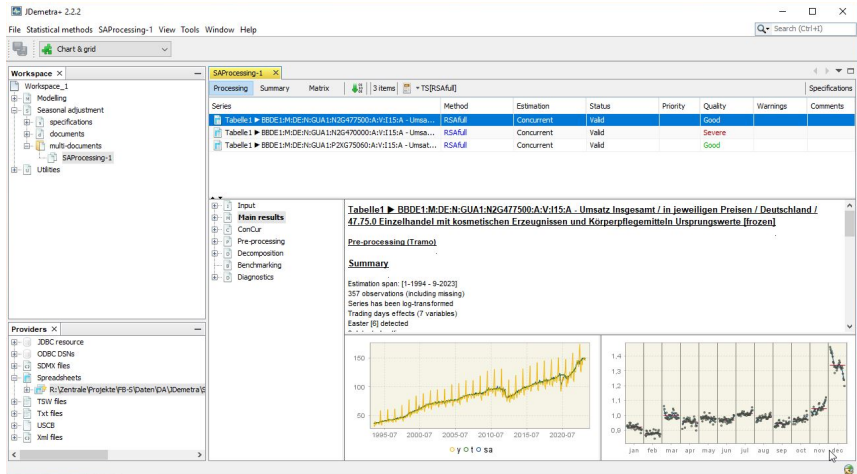
Basic functionalities

Seasonal Adjustment in JD+ (III/III)



Basic functionalities

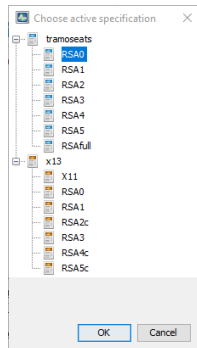
Seasonal Adjustment in JD+ (III/III)



Exercise: Seasonal adjustment

Create a new multi-document containing the previously loaded time series.

- Run the seasonal adjustment with the specification **RSaFull** for the entire workspace.
- Right-click on a time-series and select **Specification**→**Select...**
Choose a specification, click **OK** and click on the time series ("SAItem") to run the SA with the chosen specification.



Basic functionalities

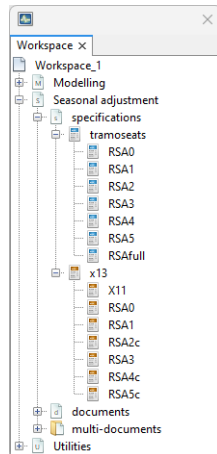
SA specifications (I/III)

tramoseats

- methods of TS

x13

- methods of X-13



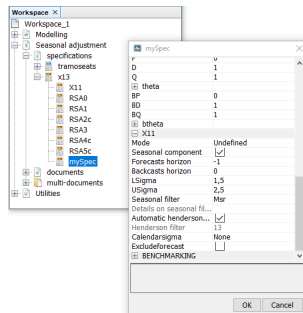
How to create a new specification:

Basic functionalities

SA specifications (II/III)

How to create a new specification:

1. Right click on a pre-defined specification
2. Select **Clone**
3. Open the new (cloned) specification

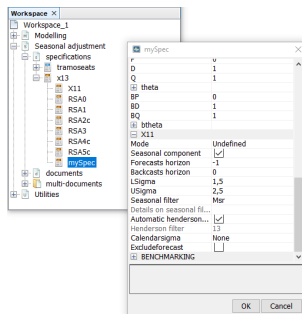


Basic functionalities

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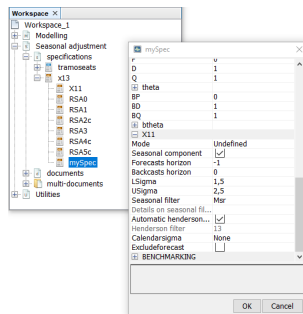


Basic functionalities

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4. Modify the settings as desired
5. Click **OK**



Basic functionalities

SA specifications (III/III)

JDemetra 2.2.2

File Statistical methods SAProcessing-1 View Tools Window Help

Chart & grid

Workspace 1

- Modeling
 - Seasonal adjustment
 - specifications
 - documents
 - mult-documents
 - SAProcessing-1
- Utilities
 - Calendars
 - Variables

Providers

- JBBC resource
- ODBC DSNs
- SDMX files
- Spreadsheets
- R: [Zentrale\Projekte\FB-S\daten\DA\JDemetra]
- TSIW files
- Txt files
- USCB
- Xml files

SAProcessing-1

Processing Summary Matrix 3 items TS[RSAsfJ]

Series	Method	Estimation	Status	Priority	Quality	Warnings	Comments
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Tabelle1 ► BBDE1:M:DE:N:GUA1:NZG470000:A-V:115:A - Umsa...	RSAsfJ	Concurrent	Valid		Severe		
Tabelle1 ► BBDE1:M:DE:N:GUA1:1P2XG75060:A-V:115:A - Umsa...	RSAsfJ	Concurrent	Valid		Good		

Input

Main results

ConCur

Pre-processing

Decomposition

Benchmarking

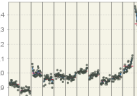
Diagnostics

Tabelle1 ► BBDE1:M:DE:N:GUA1:NZG477500:A-V:115:A - Umsatz Insgesamt / in jeweiligen Preisen / Deutschland / 47.75.0 Einzelhandel mit kosmetischen Erzeugnissen und Körperpflegemitteln Ursprungswerte [frozen]

Pre-processing (Tramo)

Summary

Estimation span: [1-1994 - 9-2023]
357 observations (including missing)
Series has been log-transformed
Trading days effects (7 variables)

Series

- Series span: All
- Preliminary Check: ☒
- ESTIMATE: ☒
- Model span: All
- Tolerance: 0,0000001
- Exact ML: ☒
- Unit root limit: 0,96
- TRANSFORMATION:
 - Function: Auto
 - Fct: 0,95
- REGRESSION:
 - Calendar:
 - Trading days: in use
 - Easter: in use
 - Option: IncludeEaster
 - Julian: ☐
 - Duration: 6
 - Test: ☒

Documented results

Apply Restore Save

Basic functionalities

SA specifications (III/III)

Apply Recalculates the selected SA-Item without saving the changed specification

☒ SERIES	
☒ ESTIMATE	
☒ TRANSFORMATION	
☒ REGRESSION	
☒ Calendar	
Pre-specified outliers	
Intervention variables	
Ramp effects	
User-defined variables	
Fixed regression coeff...	
☒ OUTLIERS	
Is enabled	☒
Detection span	All
Use default critical value	☒
Critical value	4
Additive	☒
Level shift	☒
Transitory change	☒
Seasonal outlier	☒
T/C rate	0,7
Method	AddOne
☒ ARIMA	
Automatic	☐
Mean	☐
P	0
D	1
Q	1
☒ theta	
BP	0
BD	1
BQ	1
☒ btheta	
☒ X11	
Mode	Undefined
Seasonal component	☒
Forecasts horizon	-1
Backcasts horizon	0
LSigma	1,5
USigma	2,5
Seasonal filter	Mar
☒ Details on seasonal...	
Automatic henderson f...	☒
Henderson filter	1,3
Calendarsigma	None
Excludeforecast	☐
☒ BENCHMARKING	

LSigma
[siglamin] Lower sigma boundary for the detection of extreme values.

ApplyRestoreSave

Basic functionalities

SA specifications (III/III)

Apply Recalculates the selected SA-Item without saving the changed specification

Restore Undoes all the changes since the last saved specification

☒ SERIES	
☒ ESTIMATE	
☒ TRANSFORMATION	
☒ REGRESSION	
☒ Calendar	
Pre-specified outliers	
Intervention variables	
Ramp effects	
User-defined variables	
Fixed regression coeff...	
☒ OUTLIERS	
Is enabled	☒
Detection span	All
Use default critical value	☒
Critical value	4
Additive	☒
Level shift	☒
Transitory change	☒
Seasonal outlier	☒
T/C rate	0,7
Method	AddOne
☒ ARIMA	
Automatic	☐
Mean	☐
P	0
D	1
Q	1
☒ theta	
BP	0
BD	1
BQ	1
☒ btheta	
☒ X11	
Mode	Undefined
Seasonal component	☒
Forecasts horizon	-1
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Estimate	1,5
USigma	2,5
Seasonal filter	Mar
☒ Details on seasonal...	
Automatic henderson f...	☒
Henderson filter	1,3
Calendarsigma	None
Excludeforecast	☐
☒ BENCHMARKING	

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ApplyRestoreSave

Basic functionalities

SA specifications (III/III)

Apply Recalculates the selected SA-Item without saving the changed specification

Restore Undoes all the changes since the last saved specification

Save Saves the specification and recalculates the seasonal adjustment

[+] SERIES	
[+] ESTIMATE	
[+] TRANSFORMATION	
[+] REGRESSION	
[+] Calendar	
Pre-specified outliers	
Intervention variables	
Ramp effects	
User-defined variables	
Fixed regression coeff...	
[+] OUTLIERS	
Is enabled	☒
Detection span	All
Use default critical value	☒
Critical value	4
Additive	☒
Level shift	☒
Transitory change	☒
Seasonal outlier	☒
TC rate	0,7
Method	AddOne
[+] ARIMA	
Automatic	☐
Mean	☐
P	0
D	1
Q	1
[+] theta	
BP	0
BD	1
BQ	1
[+] btheta	
[+] X11	
Mode	Undefined
Seasonal component	☒
Forecasts horizon	-1
Backcasts horizon	0
Estimate	☒
USigma	2,5
Seasonal filter	Mar
[+] Details on seasonal...	
Automatic henderson f...	☒
Henderson filter	1,3
Calendarsigma	None
Excludeforecast	☐
[+] BENCHMARKING	
[LSigma	
[sigma]min] Lower sigma boundary for the detection of extreme values.	
Apply	Restore
Save	

Exercise: Specifications

- The time series

BBDE1.M.DE.W.GUA1.N2G470000.A.V.I15.A

is the calendar adjusted figure corresponding to

BBDE1.M.DE.N.GUA1.N2G470000.A.V.I15.A

Can you alter the later one's specification to improve the seasonal adjustment's quality from "Severe" to "Good"?

Basic functionalities

Saving the current work (I/III)

Workspace Concept for gathering the current workload and making it reusable at a later time and for others.

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Workspace Concept for gathering the current workload and making it reusable at a later time and for others.

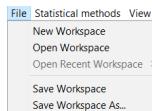
- Only a single workspace can be opened.
- A **JD+** workspace contains all specifications, multi-documents, variables and time series.
- It consists of a superior xml file and a directory of the same name with subdirectories containing xml files.

Workspace Concept for gathering the current workload and making it reusable at a later time and for others.

- Only a single workspace can be opened.
- A **JD+** workspace contains all specifications, multi-documents, variables and time series.
- It consists of a superior xml file and a directory of the same name with subdirectories containing xml files.
- Therefore a **JD+** workspace is exchangeable between different users.

Basic functionalities

Saving the current work (II/III)



New Workspace

Opens a new empty workspace

Open Workspace

Opens an existing workspace by selecting its superior xml file

Open Recent Workspace

Displays a list of previously opened workspaces to be opened

Save Workspace

Saves the current workspace (create/overwrite)

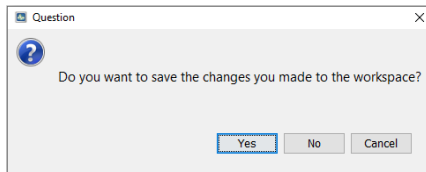
Save Workspace As...

Saves the current workspace under a selectable name in a selectable location

Basic functionalities

Saving the current work (III/III)

If you change to another workspace or the current workspace or the application is closed, JD+ will let you save the workspace.



Exercise: Workspace II

Create a subdirectory JD+/Tuesday in your local drive

C:/Daten

and save your workspace there.

■ Agenda

- Basic functionalities in JD+
- Enhanced functionalities via plug-ins
- Summary and useful links

Plug-ins

Infinite possibilities of extensions with plug-ins

It is possible to extend the application with the installation of a plug-in (an add-in which integrates new features in the existing application)

Possible examples:

- New seasonal adjustment
- Additional data provider
- Customize the application to your own needs and style
- New features what you can do with a time series

Enhanced functionalities via plug-ins

How to install a plug-in?

How to install a plug-in

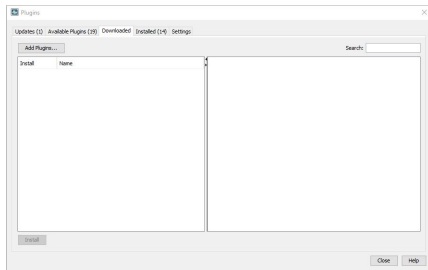
1. Download the plug-in as an nbm-file

Enhanced functionalities via plug-ins

How to install a plug-in?

How to install a plug-in

1. Download the plug-in as an nbm-file
2. Open the plug-in window in the menu bar *Tools* → *Plugins*

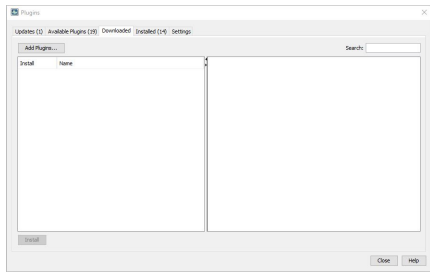


Enhanced functionalities via plug-ins

How to install a plug-in?

How to install a plug-in

1. Download the plug-in as an nbm-file
2. Open the plug-in window in the menu bar *Tools* → *Plugins*
3. Drag'n'drop the nbm-file in the window on the "*Downloaded*" tabpage **OR** select the file by clicking "*Add Plugins...*"

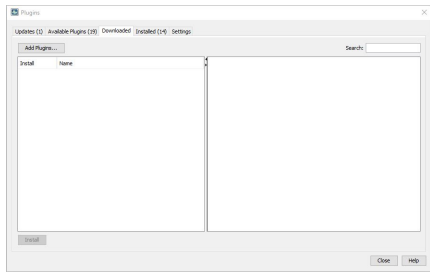


Enhanced functionalities via plug-ins

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4. Click on the button "*Install*" and follow the instructions



Exercise: Plug-ins

Install the following plug-ins, nbm-files available in the exchange directory

R:/Zentrale/3/TZK/Daten/INT_SeasonalAdjustment/02Tuesday/JD+KeyCap

TransReg Transformation of regression variables,
[Link to GitHub](#)

ConCur Controlled Current, [Link to GitHub](#)

ConCurReport Controlled Current with html output,
[Link to GitHub](#)

ExcelAdapter Excel connection to ConCur, [Link to GitHub](#)

DataProviderRandom Generates random data for an ARIMA model
of your choice

Xlsx2Ws Convert parts of Workspaces to xlsx files
and back, [Link to GitHub](#)

■ Agenda

- Basic functionalities in JD+
- Enhanced functionalities via plug-ins
- Summary and useful links

Summary

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- load time series into JD+

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Summary You know now how to...

- load time series into JD+
- process seasonal adjustment in JD+
- individualize the SA specifications for your needs
- save and share your settings and results
- enhance the base application JD+ with plug-ins

Useful links I

Release 2.2.6

 <https://github.com/jdemetra/jdemetra-app/releases/tag/v2.2.6>
 ~ Application

 <https://github.com/jdemetra/jwsacruncher/releases> ~ Cruncher

Documentation

 <https://jdemetra-new-documentation.netlify.app/>

Helpdesk (Eurostat CROS-Portal)

 <https://.../help-jdemetra-and-time-series-analysis>

Source Code (Open Source)

 <https://github.com/jdemetra/jdemetra-app>

 <https://github.com/jdemetra/jdemetra-core>

Plug-ins

 <https://github.com/bbkrd/TransReg>

 <https://github.com/bbkrd/CompRes>

 <https://github.com/bbkrd/KIX-UI>,
<https://github.com/bbkrd/KIX2.0>

Documentation

 <https://bbkrd.github.io/>

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